LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	28
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Typical cabin pressurization system, Cabin pressure regulator safety v/v, outflow v/v etc and its operation, Methods of sealing of cabin
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Typical cabin pressurization system, Cabin pressure regulator safety v/v, outflow v/v etc and its operation, Methods of sealing of cabin
5. Specific Objectives	23.2.2.1Trainee will be able to understand the necessity of cabin pressurization and working principles of components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of system diagrams and component functions; Instructor supervision of trainee identification of regulator and outflow valve roles
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Why is cabin pressurization necessary for high-altitude flight?
(b) What are the primary components of a typical cabin pressurization system?
(c) How does the cabin pressure regulator control the cabin altitude?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Necessity of cabin pressurization • Working principles of pressurization system • Cabin pressure regulator safety valve operation • Outflow valve operation and control • Methods of sealing the cabin	L & Ds	PPT, IB	15 min
(b)	• Detailed analysis of regulator safety valve • Outflow valve mechanical operation • Cabin sealing techniques and materials • System interconnections and flow paths • Emergency depressurization procedures	L & Ds	PPT, PDF	20 min
(c)	• Review of component functions • Operational sequence of pressurization • Safety mechanisms in the system • Sealing integrity verification methods • Summary of system operation	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the function of the cabin pressure regulator safety valve.
(b) Describe the operation of the outflow valve during normal and emergency conditions.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between the air conditioning system and the cabin pressurization system, referencing LP 29.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Typical air conditioning system, various components, types of systems,	Refer to the master syllabus reference list	LP 29

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min